

Lab Task – 1

Reinforcement Learning Laboratory

Experiment 1: Exploring Reinforcement Learning Environments using Gymnasium

Duration: 3 Hours

Course Outcome: CO1

Task 1: Environment Setup

- Install Gymnasium, NumPy, and Matplotlib.
- Import the required libraries.
- Verify the installation by printing the Gymnasium version.

Expected Output: Successful installation and Gymnasium version displayed.

Task 2: Create and Initialize an RL Environment

- Create the CartPole-v1 environment.
- Reset the environment.
- Display the initial observation and environment information.

Task 3: Explore Observation and Action Spaces

- Display the observation space.
- Display the action space.
- Display the type of observation space.
- Display the number of possible actions.

Task 4: Execute a Random Agent

- Execute a random action at every time step.
- Run the environment for one complete episode.
- Print step number, selected action, observation, reward, and episode status.
- Display the total number of steps and cumulative reward.

Expected Learning Outcome

Students will be able to configure Gymnasium, create and interact with RL environments, interpret observation and action spaces, execute a random agent, analyze rewards and episode termination, and compare standard RL benchmark environments.